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Reading: E&T {4.3-4.3.3}

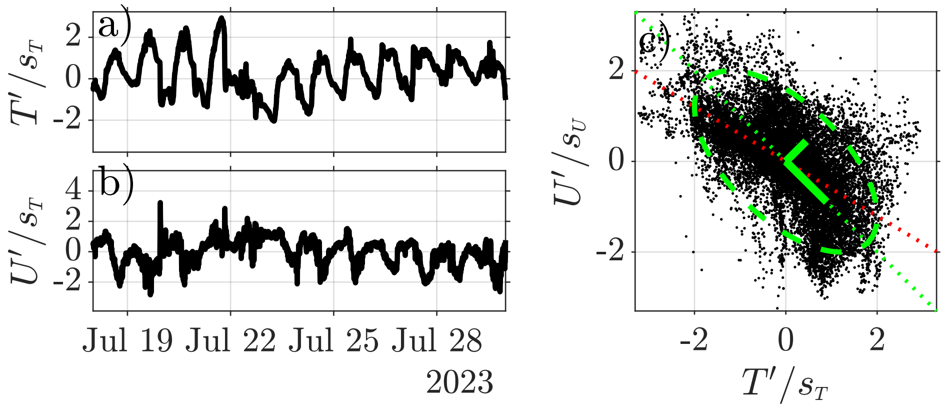

Figure 1: Time series of CMS-pier standard normalized a) air temperature $(T - \bar{T})/s_T$ and b) east/west wind speed $(U - \bar{U})/s_U$ versus time. c) Scatter diagram of U'/s_U versus T'/s_T (black) with lines indicating the correlation coefficient $\hat{\rho}_{U,T} = -0.6$ (thin dotted red), a perfect negative correlation $\rho = -1$ (thin dotted greed) and the orthogonal vectors (EOFs, $\hat{\phi}_k$) scaled by the corresponding $\hat{\sigma}_k$ (solid green).

Data Decomposition

When analyzing multiple data-sets that exhibit patterns of co-variation it is desirable to reduce the complexity (or dimensionality) of the data. That is, extract the co-varying portion of the signal and the corresponding pattern between sensors. For example, consider the set of time-series data consisting of the CMS-pier perturbation temperature $x(t_n) = T - \bar{T}$ and east-wind speed $y(t_n) = U - \bar{U}$. We have already seen that these two records are correlated, with $\hat{\rho}_{U,T} = -0.6$, and coherent in the diurnal frequency band, with $\gamma^2_{U,T} = 0.9$ and phase of $\hat{\phi}_{U,T} \approx 153°$. As a hypothetical decomposition, we might separate the coherent portion of x and y from the incoherent portion, *e.g.,*

$$y \approx \mathcal{F}(\xi) + \zeta_y,$$

where the linear function $\mathcal{F}(\xi)$ maps coherent variations in x to y and ζ_y is independent of x. A similar decomposition of x,

$$x \approx \xi + \zeta_x,$$

also results in an independent component ζ_x. With such a decomposition, we can analyze ξ and $\mathcal{F}(\xi)$ in relation to solar forcing or evaluate against a physical model for the diurnal sea-breeze. Or, analyze the parts ζ_x and ζ_y, separately, to relate them to other processes.

An efficient method for such analysis that extends well to an arbitrary number of sensors uses an orthogonal mode decomposition, called empirical orthogonal function (EOF) analysis or principal component analysis (PCA). The decomposition seeks patterns between sensors that maximize the amount of variance described by each pattern. For example, in the case of CMS-pier perturbation temperature and east-wind speed, we can expect the dominant "pattern" to be (warm temperature, onshore wind) and it's reciprocal, (cold temperature, offshore wind). The analysis is performed on the perturbation data matrix,

$$d_{n,m} = [x(t_n), y(t_n), \dots], \quad \text{or using matrix notation,} \quad \boldsymbol{d} = [\boldsymbol{x}, \boldsymbol{y}],$$

such that $d_{n,1} = x_n$ and $d_{n,2} = y_n$. It is important to remove the mean and trend from the data,

```
% t  = time [days] datetime fromat
% dt = sample interval [days]
```

```
% x   = temperature record [C];
% y   = East(+)/West(-) wind speed [m/s];
% N   = number of samples
%
% create data matrix
d = [x,y];
% estimate trend
tt = [0*x+1, convertTo(time,'datenum')];
bb = tt\d;
% remove the mean & linear trend
d     = d-tt*bb;
```

The decomposition of data matrix $\boldsymbol{d}$ is,

$$d_{n,m} = \sum_{k=1}^{K} a_{n,k}\phi_{m,k} + \epsilon_m, \quad \text{or in matrix form,} \quad \boldsymbol{d} = \boldsymbol{A}\boldsymbol{\Phi}^T + \boldsymbol{\epsilon} \tag{1}$$

with $\phi_{m,k}$ quantifying the pattern between x and y and amplitudes $a_{n,k}$ capturing time variations of each pattern. The columns of $\boldsymbol{A}$ and $\boldsymbol{\Phi}$ contain the temporal amplitude $\boldsymbol{a}_k = [a_{1,k}, \ldots, a_{N,k}]$ and spatial pattern $\boldsymbol{\phi}_k = [\phi_{1,k}, \ldots, \phi_{M,k}]$, respectively. Note, hereafter ϕ will be used to denote modes not an angle and we will refer to the m direction as space (though it may be different sensors at the same location) and n as time. We will see later that the maximum number of modes K is equivalent to the number of sensors, say M. When the maximum number of modes are used $K = M$, the error is zero $\epsilon \approx 0$ to within round-off error. In the case of the CMS-pier observations, the two modes are 2D-vectors: ϕ_1 which points south-east and ϕ_2 that points north-east (green, Figure 1c). The length of the vectors have been scaled by the standard-amplitude of each mode, σ_k, where the variance in each mode is,

$$\sigma_k^2 = \frac{1}{N}\sum_{n=1}^{N} a_{n,k}^2 = \overline{\boldsymbol{a}_k\boldsymbol{a}_k}, \tag{2}$$

and there is more variability associated with the south-east pointing ϕ_1 mode than ϕ_2. These scaled vectors can be used to form an ellipse, with semi-major axis σ_1 along ϕ_1 and semi-minor axis σ_2 along ϕ_2 (dashed green, Figure 1c, code to follow). The ellipse can be scaled to encircle various percentages of the scattered data, *e.g.,* for normally distributed data 95% is within a radii of $2\sigma_k$. This is a rudamentary way of "de-spiking"; estimate the covariance ellipse between the data vector x and its first derivative $\dot{x}$ (and possibly second), then remove data with $(x, \dot{x})$ outside an appropriately scaled ellipse.

Normal Modes

The modes are vectors $\boldsymbol{\phi}_j = [\phi_{1,j}, \phi_{2,j}, \ldots]^T$ describing a certain pattern of covariability in space. For example, in the non-dimensional scatter plot of CMS-pier data, $\boldsymbol{\phi}_1 = [1, -1]^T/\sqrt{2}$ is closely aligned along the widest part of the data cloud and $\boldsymbol{\phi}_2 = [1, 1]^T/\sqrt{2}$ perpendicular to this (solid green, Figre 1c). So, for the mode-1 spatial pattern, a positive temperature anomaly has an associated negative wind-speed patern, and vice versa. To ensure that the patterns are distinct, and there is no spatial covariance between the modes, the vectors are required to be "orthogonal" (perpendicular to each other),

$$(\boldsymbol{\phi}_j, \boldsymbol{\phi}_k) = \boldsymbol{\phi}_j^T\boldsymbol{\phi}_k = \sum_m \phi_{m,j}\phi_{m,k} = 0, \qquad \text{if } j \neq k.$$

The factor of $\sqrt{2}$ in the CMS-pier modes normalizes them to have unit-magnitude, or form unit vectors satisfying,

$$||\boldsymbol{\phi}||^2 = (\boldsymbol{\phi}_k, \boldsymbol{\phi}_k) = \phi_{1,k}\phi_{1,k} + \phi_{2,k}\phi_{2,k} \cdots = 1. \tag{3}$$

By making the modes orthogonal, the amplitudes $a_{n,k}$ of different modes have zero temporal covariance, *i.e.,*

$$\overline{\boldsymbol{a}_j \boldsymbol{a}_k} = \frac{1}{N}(a_{1,j}a_{1,k} + a_{2,j}a_{2,k} + \dots) = 0, \qquad \text{if } j \neq k.$$

The above notation will be used moving forward. I figured it would be easiest to get it out of the way first.

Covariance Matrix Diagonalization

Before explaining how to estimate the modes and amplitudes, lets begin by estimating the (biased) data-data covariance matrix using the modal decomposition, we'll assume $\epsilon = 0$ for simplicity,

$$\begin{aligned}
C_{i,j} &= \overline{x_{n,i}x_{n,j}}, \qquad \text{averaging over the repeated subscript } n,\\
&= \overline{(\boldsymbol{a}_k\phi_{i,k})(\boldsymbol{a}_k\phi_{j,k})}, \quad \text{dropping the subscript } n\text{, and also summing over } k,\\
&= \overline{(\boldsymbol{a}_1\phi_{i,1} + \boldsymbol{a}_2\phi_{i,2} + \dots)(\boldsymbol{a}_1\phi_{j,1} + \boldsymbol{a}_1\phi_{j,2} + \dots)},\\
&= \boldsymbol{a}_1\phi_{i,1}\overline{(\boldsymbol{a}_1\phi_{j,1} + \boldsymbol{a}_1\phi_{j,2} + \dots)} + \boldsymbol{a}_2\phi_{i,2}\overline{(\boldsymbol{a}_1\phi_{j,1} + \boldsymbol{a}_1\phi_{j,2} + \dots)} + \dots,\\
&= \underbrace{\overline{\boldsymbol{a}_1\boldsymbol{a}_1}}_{\sigma_1^2}\phi_{i,1}\phi_{j,1} + \cancelto{0}{\overline{\boldsymbol{a}_1\boldsymbol{a}_2}}\phi_{i,1}\phi_{j,2} + \dots + \cancelto{0}{\overline{\boldsymbol{a}_2\boldsymbol{a}_1}}\phi_{i,2}\phi_{j,1} + \underbrace{\overline{\boldsymbol{a}_2\boldsymbol{a}_2}}_{\sigma_2^2}\phi_{i,2}\phi_{j,2} + \dots,\\
&= \sum_k \sigma_k^2\phi_{i,k}\phi_{j,k}.
\end{aligned}$$

In matrix notation the above equation is equivalent to,

$$\boldsymbol{C} = \boldsymbol{\Phi}\boldsymbol{\Lambda}\boldsymbol{\Phi}^T. \tag{4}$$

where the matrices are,

$$\boldsymbol{\Phi} = \begin{bmatrix}\boldsymbol{\phi}_1 & \dots & \boldsymbol{\phi}_M\end{bmatrix} = \begin{bmatrix}\phi_{1,1} & \dots & \phi_{1,M}\\ \vdots & \ddots & \vdots\\ \phi_{M,1} & \dots & \phi_{M,M}\end{bmatrix} \tag{5}$$

and,

$$\boldsymbol{\Lambda} = N\begin{bmatrix}\sigma_1^2 & 0 & \dots & 0\\ 0 & \sigma_2^2 & \ddots & \vdots\\ \vdots & \ddots & \ddots & 0\\ 0 & \dots & 0 & \sigma_M^2\end{bmatrix} \tag{6}$$

The procedure of going from the left-hand-side of (4) to the right-hand-side is called diagonalization, because it results in a diagonal $M \times M$ matrix $\boldsymbol{\Lambda}$, with the variance of each mode σ_k^2. The matrix $\boldsymbol{\Phi}$ contains the modes of $\boldsymbol{d}$ and there are at most M modes. By the orthogonality and unit magnitude of the vectors, the transpose $\boldsymbol{\Phi}^T$ is the inverse matrix,

$$\boldsymbol{\Phi}^T\boldsymbol{\Phi} = \boldsymbol{I},$$

where $\boldsymbol{I}$ is the identity matrix (ones on diagonal, zeros everywhere else). Thus, we can rewrite (4) as,

$$\boldsymbol{\Phi}^T\boldsymbol{C}\boldsymbol{\Phi} = \boldsymbol{\Lambda} \tag{7}$$

illustrating that the matrix $\boldsymbol{\Phi}$ rotates the covariance matrix into the diagonal matrix $\boldsymbol{\Lambda}$.

The diagonalization preserves variance. If we treat the sum of diagonal elements of $\boldsymbol{C}$, trace($\boldsymbol{C}$) $= C_{1,1} + C_{2,2} + \dots$, as the total variance of the input record, then although the diagonal elements of $\boldsymbol{\Lambda}$ will differ from $\boldsymbol{C}$ the sum is the same, *i.e.,* trace($\boldsymbol{\Lambda}$) = trace($\boldsymbol{C}$). For example, for the CMS-pier trace($\boldsymbol{C}$) $= s_x^2 + s_y^2 \approx$

$4.9\,[\mathrm{C}]^2 + 5.5\,[\mathrm{m/s}]^2$ whereas the diagonal elements of $\boldsymbol{\lambda}$ are $\sigma_1^2 = 8.3$ and $\sigma_2^2 = 2.1$. We can use this to quantify the "fraction of variance" (FOV) explained by each mode, $\sigma_k^2/\sum_j \sigma_j^2$. For the CMS-pier data, σ_1^2 and σ_2^2 account for 80% and 20% of the total variance, respectively. It is common to label modes as "noise" when the FOV is less than expected for random data. Based on a Monte-Carlo simulation of 1000 random data sets, the 95%-limits are 51% and 47% (code at end of notes). From this, we may conclude that mode-1 is significant, with mode-2 variability accounting for independent/random variations.

Eigen-decomposition of the Covariance Matrix

As an additional constraint, we will organize the diagonal matrix so that the modes containing the most variance come first, *i.e.,*

$$\sigma_1^2 \geq \sigma_2^2 \geq \cdots \geq \sigma_M^2.$$

By doing so, we can chose to reconstruct our data using a truncated sum $K < M$, *e.g.,* only mode-1 for the CMS-pier data, and are guaranteed that modes describing larger portions of the variance are retained and small portions are neglected. The procedure for estimating the mode-1 vector $\boldsymbol{\phi}_1$, amounts to minimizing the residual,

$$\epsilon_1^2 = \frac{1}{N}\sum_{n=1}^{N} ||\boldsymbol{d} - (\boldsymbol{d}, \boldsymbol{\phi}_1)\boldsymbol{\phi}_1||, \tag{8}$$

where the time-varying mode-1 amplitude $a_{n,1} = (\boldsymbol{d}, \boldsymbol{\phi}_1)$ is the projection of the data onto the mode-1 vector. The residual may be expanded and simplified using notation from above,

$$\begin{aligned}
\epsilon_1^2 &= \overline{(\boldsymbol{d}, \boldsymbol{d})} - 2\,\underbrace{\overline{(\boldsymbol{d}, a_{n,1}\boldsymbol{\phi}_1)}}_{a_{n,1}(\boldsymbol{d},\boldsymbol{\phi}_1)} + \underbrace{\overline{(a_{n,1}\boldsymbol{\phi}_1, a_{n,1}\boldsymbol{\phi}_1)}}_{a_{n,1}^2(\boldsymbol{\phi}_1,\boldsymbol{\phi}_1)}, \\
&= \mathrm{trace}(\boldsymbol{C}) - \underbrace{\overline{\boldsymbol{a}_1\boldsymbol{a}_1}}_{\sigma_1^2} \\
&= \mathrm{trace}(\boldsymbol{C}) - \boldsymbol{\phi}_1^T\boldsymbol{C}\boldsymbol{\phi}_1,
\end{aligned} \tag{9}$$

where the last two lines come from applying (7) using only the first mode, *i.e.,* $\boldsymbol{\Phi} \to \boldsymbol{\phi}_1$. Thus, minimization of the residual is equivalent to maximizing the variance σ_1^2. The minimization technique is similar to estimating least-squares coefficients, except that the normalization constraint (3) is included in the minimization via Lagrange multiplier λ, *i.e.,*

$$\begin{aligned}
\frac{\partial}{\partial\boldsymbol{\phi}_1}\left(\epsilon_1^2 + \lambda(\boldsymbol{\phi}^T\boldsymbol{\phi} - 1)\right) &= 0, \\
-2\boldsymbol{C}\boldsymbol{\phi}_1 + 2\lambda\boldsymbol{\phi}_1 &= 0, \\
\boldsymbol{C}\boldsymbol{\phi}_1 &= \lambda\boldsymbol{\phi}_1,
\end{aligned} \tag{10}$$

which is the standard eigenvalue problem. There are at most M solutions, determined by the roots of the characteristic polynomial,

$$\det|\boldsymbol{C} - \boldsymbol{I}\lambda| = 0.$$

For an $M = 2$ data-set, like the CMS-pier observations, the determinant is,

$$\det\begin{bmatrix} (s_x^2 - \lambda) & s_{x,y}^2 \\ s_{x,y}^2 & (s_y^2 - \lambda) \end{bmatrix} = \lambda^2 - (s_x^2 + s_y^2)\lambda + (s_x^2 s_y^2 - s_{x,y}^4), \tag{11}$$

and the roots of the characteristic polynomial are,

$$\lambda^{\pm} = \frac{(s_x^2 + s_y^2)}{2} \pm \sqrt{\left(\frac{s_x^2 - s_y^2}{2}\right)^2 + s_{x,y}^4}.$$

In the event that the variances are equal, $s_x^2 = s_y^2 = s^2$, the eigenvalues simplify to,

$$\lambda^{\pm} = s^2 \pm |s_{x,y}|^2 = s^2(1 \pm |\rho_{x,y}|),$$

where, $\rho_{x,y} = s_{x,y}^2/s^2$ is the correlation coefficient. In order to maximize the fraction of variance, we select the largest eigenvalue $\lambda^+ = \sigma_1^2$ as our first mode.

The eigenvector can then be determined by solving (10) for the vector components $\boldsymbol{\phi}_1 = [\mathrm{b}_1, \mathrm{b}_2, \ldots]^T$, using the largest λ, *e.g.,* in the 2×2 case with equal variance s^2,

$$[\, s^2 \quad s_{x,y}^2 \,] \begin{bmatrix} \mathrm{b}_1 \\ \mathrm{b}_2 \end{bmatrix} = \lambda^+ \mathrm{b}_1, \tag{12}$$

which simplifies to $\mathrm{b}_1 = \mathrm{sgn}(\rho_{x,y})\mathrm{b}_2$, where the sign function $\mathrm{sgn(a)} = +1$ if argument $\mathrm{a} > 0$ and $= -1$ if $\mathrm{a} < 0$. Using the normalization constraint, $\mathrm{b}_1^2 + \mathrm{b}_2^2 = 1$, gives,

$$\boldsymbol{\phi}_1 = \begin{bmatrix} \mathrm{b}_1 \\ \mathrm{b}_2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ \mathrm{sgn}(\rho_{x,y}) \end{bmatrix}. \tag{13}$$

The mode-2 eigenvector is determined in a similar manner, substituting the next largest eigenvalue. Most software packages have an eigenvalue solver that performs this analysis efficiently. Matlab, has a function called eig() that outputs the eigen vectors and eigenvalues, but they are not ordered by magnitude, so you'll have to sort them, for example,

```
% 1) using the covariance matrix
C = (d'*d)/N;
% estimate principal components from eigenvalue decomp
[V    ,L    ] = eig(C);
% eig does not always sort the values/vectors, we want largest first
[L    ,ind] = sort(diag(L),'descend');
V = V(:,ind);
% make "x" component positive
V = V.*sign(V(1,:));
% L = total variance described by each mode
% V = EOFs
```

You'll notice that I also multiplied the vectors by the sign of the first coordinate so that the x-component of the unit vectors would be positive. The time-varying amplitude $a_{n,k}$ of the k^{th} mode are formed by projecting the original data onto the mode,

$$a_{n,k} = \sum_m d_{n,m}\phi_{m,k}, \qquad \text{or in vector form,} \qquad \boldsymbol{a}_k = \boldsymbol{d}\,\boldsymbol{\phi}_k.$$

Singular Value Decomposition

The diagonalization of the $M \times M$ covariance matrix $\boldsymbol{C}$ can be extended to non-square matrices, like our $N \times M$ data-matrix $\boldsymbol{d}$, via the singular value decomposition,

$$\boldsymbol{d} = \boldsymbol{U}\boldsymbol{S}\boldsymbol{V}^T, \tag{14}$$

where $\boldsymbol{U}$ is $N \times M$, $\boldsymbol{S}$ is a diagonal $M \times M$ matrix, and $\boldsymbol{V}$ is an $M \times M$ orthogonal matrix. The eigen-modes are $\boldsymbol{\Phi} = \boldsymbol{V}$. The eigenvalues are determined from the diagonal elements of $\boldsymbol{S}$,

$$\lambda_k = \sigma_k^2 = S_{k,k}^2/N, \qquad \text{or,} \qquad \boldsymbol{\Lambda} = \boldsymbol{S}^T\boldsymbol{S}/N.$$

The columns of $\boldsymbol{U}$ are the unit-magnitude time-varying amplitudes of each mode. Since the amplitudes of each mode are uncorrelated, the columns are also orthogonal $\boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}$. The appropriately scaled, time-varying amplitude matrix is given by the product,

$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{S},$$

the columns of $\boldsymbol{A}$ are sometimes called expansion coefficients or principal components. The k^{th} expansion coefficient is,

$$\boldsymbol{a}_k = \boldsymbol{U}_k S_{k,k}.$$

Matlab also has a routine for calculating the singular value decomposition, svd(), which is more efficient than first estimating the covariance matrix and then conducting an eigen-decomposition. Conveniently, it does sort the modes:

```
% 2) via singular value decomp
[U,S,EOFs] = svd(d,0);
% L = S^2/N is the variance associated with each mode
% A = U*S  is the amplitude (EOF coefficients)
```

Now that we have estimated the modes ϕ_k and the associated variance σ_k^2, we can construct a variance ellipse to accompany the data-data scatter plot (green dashed circle, Figure 1). The procedure is to: first, create an ellipse in the (ϕ_1,ϕ_2) coordinate system with semi-major axis σ_1 and semi-minor axis σ_2, then rotate it into the data coordinates by left multiplying the ellipse coordinates by $\boldsymbol{\Phi}$ (V in below code) or right-multiplying by $\boldsymbol{\Phi}^T$. In MATLAB code,

```
% V = [phi_1      , phi_2      ];        % EOF matrix, columns are modes
% L = [sigma_1^2, sigma_2^2];            % vector w/ variance of each mode
% estimate covariance ellipse in modal coordiantes
theta  = [0:0.01:1]*2*pi;                % angle goes 0-2\pi
p1     = 2*sqrt(L(1))*cos(theta);        % semi-major axis = 2\sigma_1
p2     = 2*sqrt(L(2))*sin(theta);        % semi-minor axis = 2\sigma_2
% rotate using EOFs to data coordiantes
xyErot = V*[p1;p2];                      % multiply modes V by amplitudes (p1,p2)
xE     = xyErot(1,:);                    % extract x-component
yE     = xyErot(2,:);                    % extract y-component
```

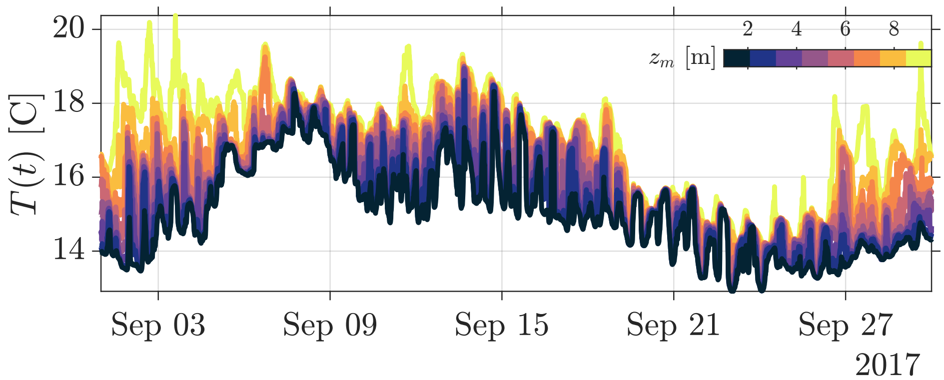

Figure 2: Time series of temperature measurements at varying distances above bottom z_m (colors).

Uncertainty

The orthogonal mode decomposition does not guarantee that individual modes (or patterns) are distinct, that is, it is possible for two (or more) modes to be "mixed". In the case of perfect measurements, it is possible for two (or more) eigenvectors $\boldsymbol{\phi}$ to share the same eigenvalue λ. Such an eigenvalue is called "degenerate". In the case of EOFs, it indicates that each corresponding eigenvector describe an equivalent fraction of variance. It also means that the actual underlying pattern may be from some combination of the two eigenvectors. For example, say both $\boldsymbol{\phi}_a$ and $\boldsymbol{\phi}_b$ are orthogonal eigenvectors of the same eigenvalue $\lambda_{a,b} = \sigma_{a,b}^2$, then any linear combination of these two vectors $\boldsymbol{\phi}_{a,b} = \mathrm{a}\boldsymbol{\phi}_a + \mathrm{b}\boldsymbol{\phi}_b$ is also an eigenvector of $\boldsymbol{C}$

with eigenvalue $\lambda_{a,b}$,

$$\begin{aligned}
\boldsymbol{C}\boldsymbol{\phi}_{a,b} &= \boldsymbol{C}(\mathrm{a}\boldsymbol{\phi}_a + \mathrm{b}\boldsymbol{\phi}_b) \\
&= \mathrm{a}\boldsymbol{C}\boldsymbol{\phi}_a + \mathrm{b}\boldsymbol{C}\boldsymbol{\phi}_b \\
&= \mathrm{a}\lambda_{a,b}\boldsymbol{\phi}_a + \mathrm{b}\lambda_{a,b}\boldsymbol{\phi}_b \\
&= \lambda_{a,b}\boldsymbol{\phi}_{a,b}.
\end{aligned} \tag{15}$$

As such, patterns encoded into $\boldsymbol{\phi}_a$ and $\boldsymbol{\phi}_b$ are ambiguous and mixtures of the two are as likely to be the true signal.

In the case of real data, it is unlikely to get degenerate eigenvalues because measurement errors and noise inject just enough randomness that truly degenerate eigenvalues appear distinct. That is, if you find two eigenvectors λ_a and λ_b, how do you decide whether they are distinct or only differ due to errors/noise? North et al., 1982[1] determined that the sampling error $\Delta\lambda$ for an eigenvalue is to leading order,

$$\Delta\lambda_k \sim \lambda_k\sqrt{\frac{2}{N_{\text{eff}}}},$$

where I use the effective degrees of freedom $N_{\text{eff}} = N/N_d$ because most oceanographic measurements have a rather long decorrelation time $N_d\Delta t$ (having a red power spectral density). In the event that two eigenvalues overlap,*e.g.,* $\lambda_a \in \lambda_b \pm \Delta\lambda_b$, then these should be considered degenerate and evaluated together. The eigenvectors are also affected by errors, to leading order,

$$\Delta\boldsymbol{\phi}_k \sim \frac{\Delta\lambda_k}{|\lambda_j - \lambda_k|}\boldsymbol{\phi}_k,$$

where λ_j is the nearest magnitude eigenvalue to λ_k. The important point here is that the closer the two eigenvalues become the larger the uncertainty in the eigenvector. In the event of degeneracy, our uncertainty is infinite!

Example: Vertical Temperature Measurements

As an example, lets apply the EOF analysis to temperature measurements from thermistors located at different water depths from the Point Sal, CA region. These data were collected as part of an Office of Naval Research funded Inner-shelf Dynamics Experiment[2] and the data and following analysis follow Feddersen et al., 2018[3]. There were eight thermistors at $z_m = \{1.6, 2.7, 3.9, 5, 6.2, 7.3, 8.5, 9\}$ meters above the bottom. The $z_m = 9$ m sensor was just below the surface and moved with the tide, the others were at a fixed distance above bottom. Temperature variations at this site exhibit several patterns, two of which are discernible from the time-series (Figure 2):

i) temperature rises and falls at all depths on time-scales of 12-24 hours,
ii) temperature difference between sensors rises and falls on time-scales ranging from 24 hours to days/weeks.

Analyzing each of these time-series separately would result in redundant information due to the patterns of covariance. We can reduce the size of our data-set by isolating the dominant patterns and analyzing them separately, á la EOFs. Applying the singular value decomposition to the data matrix,

$$\boldsymbol{T} = \begin{bmatrix}\boldsymbol{T}_1 & \boldsymbol{T}_2 & \dots & \boldsymbol{T}_8\end{bmatrix} = (\boldsymbol{U}\,\boldsymbol{S})\boldsymbol{\Phi}^T = \boldsymbol{A}\,\boldsymbol{\Phi}^T, \tag{16}$$

results in a set of eight modes $\{\boldsymbol{\phi}_1, \dots, \boldsymbol{\phi}_8\}$, modal amplitudes $\{\boldsymbol{a}_1, \dots, \boldsymbol{a}_8\}$, and corresponding variances $\{\sigma_1^2, \dots, \sigma_8^2\}$.

[1] `https://doi.org/10.1175/1520-0493(1982)110%3C0699:SEITEO%3E2.0.CO;2`
[2] `https://doi.org/10.1175/BAMS-D-19-0281.1`
[3] `https://doi.org/10.1029/2019JC015347`

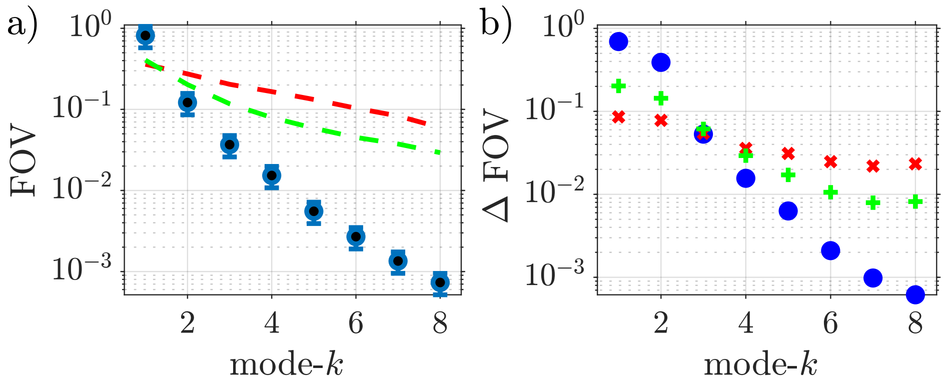

Figure 3: a) Fraction of variance FOV, $\sigma_k^2/\text{trace}(\boldsymbol{C})$, versus mode-$k$ for the temperature observations (blue), with uncertainties $\Delta\lambda_k$ shown as error bars. Also, shown are noise floor FOV estimates based on Monte-Carlo simulations of random data (dashed red) and random shuffles of the data vectors (dashed green). b) Difference in FOV between neighboring modes (same colors).

The first analysis step is to estimate the uncertainties of each eigenvalue, $\Delta\lambda_k = \sigma_k^2\sqrt{2/N_{\text{eff}}}$, and the noise floor for statistical significance relative to random noise λ_{ϵ_k}. The depth-averaged decorrelation time $\tau = 2$ day giving $N_{\text{eff}} = 23$, or $\sqrt{2/N_{\text{eff}}} = 0.3$. The fraction of variance (FOV) is dominated by mode-1, which accounts for 80% of the variance (blue, Figure 4a). The second mode $k = 2$ accounts for only 12% of the variance, deciding whether this is "significant" is tricky due to how much variance was explained by mode-1. We can perform a Monte-Carlo simulation by generating 1000 surrogate data-sets composed of independent normally distributed variables and estimate the 95% confidence limits,

```
Ne   = 23;              % effective degrees of freedom
Ni   = 1e3;             % number of iterations
M    = 8;               % number of sensors
F    = nan(M,Ni);     % log for fraction of variance
for  ii=1:Ni
     x   = randn(round(Ne),M); % random data matrix
     [~,S,~] = svd(x);          % singular value decomp on x
     L   = diag(S).^2/Ne;       % estimate variance of each mode
     F(:,ii) = L/sum(L);        % log the fraction of variance
end
F   = sort(F');                  % sort variance in ascending order
CI = F(round(0.95)*Ni,:); % find the 95% maximum value
```

The 95%-CI FOV explained by mode-1 and mode-2 are 0.4 and 0.25, respectively. Based on this, we might conclude that modes 2-8 are no more significant than those from independent random data. However, this does not account for the significant portion of the variance explained by mode-1. As an alternative to comparing the FOV, we can look at the difference in FOV, ΔFOV, between neighboring modes (Figure 4b). In doing so, and comparing FOV to ΔFOV it is apparent that random data has variance distributed more uniformly all across modes (small slopes in Figure 4a-b), relative to our observations. For the observations, ΔFOV for mode-2 and mode-3 is larger than for random data (mode-3 is borderline!). More sophisticated error analyses exist, *e.g.,* randomly shuffle the data on time-scales comparable to the decorrelation scale, or dominant time-scales like tides or seasons (*e.g.,* Peng & Fyfe 1996)[4], and at each iteration estimate σ_k^2 and logging the 95% CI for each σ_k^2. The results of one such simulation are shown in green in Figure 4a-b. In this case, the ΔFOV for the observed mode-3 is below the randomly shuffled CI, suggesting that it may not be significant. With this, it is safe to conclude that mode-1 and mode-2 are statistically significant, which account for 93.8% of the variability. Not bad!

The EOF-analysis has significantly reduced the complexity of our data. Instead of analyzing 8 time-series to relate to forcing/dynamics, we can analyze two time-series $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ associated with the dominant patterns

[4] `://doi.org/10.1175/1520-0442(1996)009%3C1824:TCPBSL%3E2.0.CO;2~`

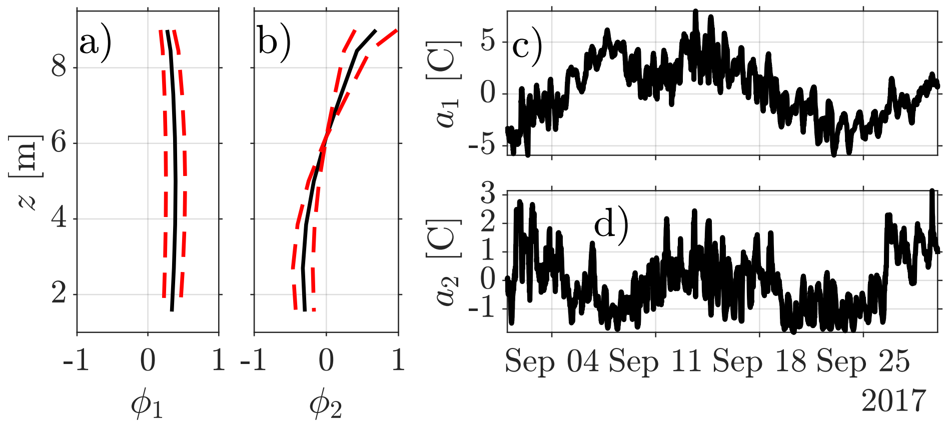

Figure 4: Temperature empirical orthogonal function vertical structure of a) mode-1 and b) mode-2 and corresponding temporal amplitudes for c) mode-1 and d) mode-2. In (a)-(b), the uncertainty $\Delta\boldsymbol{\phi}_k$ are included as red-dashed lines.

in $\boldsymbol{\phi}_1$ and $\boldsymbol{\phi}_2$.

The principal component analysis of CMS-pier observations in Figure 1c was done using the following code:

```
% t  = time [days] datetime fromat
% dt = sample interval [days]
% x  = temperature record [C];
% y  = East(+)/West(-) wind speed [m/s];
% N  = number of samples
%
% create data matrix
d = [x,y];
% estimate trend
tt = [0*x+1, convertTo(time,'datenum')];
bb = tt\d;
% remove the mean & linear trend
d   = d-tt*bb;
s2  = var(d);
d   = d./sqrt(s2);
%
% EOFs two ways:
% 1) using the covariance matrix
C = (d'*d)/N;
% estimate principal components from eigenvalue decomp
[V    ,L   ] = eig(C);
% eig does not always sort the values/vectors, we want largest first
[L    ,ind] = sort(diag(L),'descend');
V = V(:,ind);
% make "x" component positive
V = V.*sign(V(1,:));
% L = total variance described by each mode
% V = EOFs
%
% 2) via singular value decomp
[U,S,EOFs] = svd(d,0);
% L = S^2/N is the variance associated with each mode
% A = U*S  is the amplitude (EOF coefficients)
%
```

```
% generate a figure and axes
xm  = 2; ym  = 2; pw  = 12; ph  = 4; ag=0.5;
ppos0=[xm              ym                    3/2*ph-ag  (ph-ag/2)/2];
ppos1=[xm              ym+(ph-ag/2)/2+ag/2   3/2*ph-ag  (ph-ag/2)/2];
ppos2=[xm+3*ph/2+2.5*ag ym                   ph         ph];
ps   =[2*xm+9*ph/4+ag  1.25*ym+ph];
%
% create figure
fig1= figure('units','centimeters','color','w');
pos = get(fig1,'position');
pos(3:4)=ps;
set(fig1,'position',pos,'papersize',ps,'paperposition',[0 0 ps])
% create time-series axes: temperature
f1ax0 = axes('units','centimeters','position',ppos1);
f1ax0p1=plot(f1ax0,time,d(:,1),'-k','linewidth',2);
ylabel(f1ax0,'$T''/s_{\scriptscriptstyle T}$','interpreter','latex')
text(f1ax0,0.01,0.9,'a)','interpreter','latex','units','normalized')
grid on
set(f1ax0,'tickdir','out','ticklabelinterpreter','latex',...
        'xlim',tlimits,'ylim',1.1*[min(d(:,1)) max(d(:,1))],'xticklabel',[])
% create time-series axes: wind-speed
f1ax1 = axes('units','centimeters','position',ppos0);
f1ax1p1=plot(f1ax1,time,d(:,2),'-k','linewidth',2);
ylabel(f1ax1,'$U''/s_{\scriptscriptstyle T}$','interpreter','latex')
text(f1ax1,0.01,0.9,'b)','interpreter','latex','units','normalized')
grid on
set(f1ax1,'tickdir','out','ticklabelinterpreter','latex',...
        'xlim',tlimits,'ylim',1.1*[min(d(:,2)) max(d(:,2))])
%
% create scatter plot axes: Wind vs Temperature
f1ax2  = axes('units','centimeters','position',ppos2);
f1ax2p1=plot(f1ax2,d(:,1),d(:,2),'.k',...
            [min(d(:)) max(d(:))],[min(d(:)) max(d(:))]*C(2)/sqrt(prod(diag(C
              ))),':r',...
            [min(d(:)) max(d(:))],[min(d(:)) max(d(:))]*(-1),':g',...
            'MarkerSize',0.5);
hold on,
f1ax2p3=plot([zeros(1,M);(sqrt(L).*EOFs(1,:)')'],[zeros(1,M);(sqrt(L).*EOFs
   (2,:)')'],'-','color','g','linewidth',3);
xlabel(f1ax2,'$T''/s_{\scriptscriptstyle T}$','interpreter','latex')
ylabel(f1ax2,'$U''/s_{\scriptscriptstyle U}$','interpreter','latex')
text(f1ax2,0.01,0.9,'c)','interpreter','latex','units','normalized')
grid on
set(f1ax2,'tickdir','out','ticklabelinterpreter','latex',...
        'xlim',1.1*[-3 3],'ylim',1.1*[-3 3])
```